

```
In [1]: !pip install fg-data-profiling pandas numpy --quiet
```

```
In [2]: import pandas as pd
import numpy as np
import data_profiling
from data_profiling import ProfileReport
```

```
In [23]: #Loading data set to be reviewed and cleaned
df = pd.read_csv('My_Projects/Urban_Forestry/Cityworks_SR/Cityworks_SRs_2025.csv')
```

```
In [24]: #Generating an initial Profile Report of the data set to obtain a baseline understanding of the structure
profile = ProfileReport(df, title="Data Profile Report", explorative=True)
profile.to_notebook_iframe()
```

```
Summarize dataset: 0%|          | 0/5 [00:00<?, ?it/s]
```

```
0%|          | 0/14 [00:00<?, ?it/s]
14%|          | 2/14 [00:00<00:02, 5.16it/s]
100%|          | 14/14 [00:03<00:00, 4.22it/s]
```

```
Generate report structure: 0%|          | 0/1 [00:00<?, ?it/s]
```

```
Render HTML: 0%|          | 0/1 [00:00<?, ?it/s]
```

Data Profile Report



Overview

Overview

[Alerts](#) 15[Reproduction](#)

Dataset statistics

Number of variables	14
Number of observations	5893
Missing cells	376
Missing cells (%)	0.5%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	4.4 MiB
Average record size in memory	774.3 B

Variable types

Numeric	1
Categorical	9
DateTime	1
Text	3

Variables

[Select Columns](#)

RequestId

In [25]: *#Printing another structural overview of the data set*
`df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5893 entries, 0 to 5892
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   RequestId             5893 non-null   int64
1   Problem Code          5893 non-null   object
2   Date Initiated        5893 non-null   object
3   Description            5893 non-null   object
4   Priority               5890 non-null   float64
5   Category              5890 non-null   object
6   Submit To             5816 non-null   object
7   Dispatch To           5889 non-null   object
8   Address                5890 non-null   object
9   Zone                  5881 non-null   object
10  Council District      5629 non-null   object
11  Keymap                5890 non-null   object
12  Zip Code              5889 non-null   object
13  Status                5890 non-null   object
dtypes: float64(1), int64(1), object(12)
memory usage: 644.7+ KB
```

In [26]: *#Calculating the sum of all missing values for each column*
`df.isna().sum()`

```
Out[26]: RequestId      0
        Problem Code    0
        Date Initiated  0
        Description     0
        Priority         3
        Category        3
        Submit To       77
        Dispatch To     4
        Address         3
        Zone            12
        Council District 264
        Keymap          3
        Zip Code        4
        Status          3
        dtype: int64
```

```
In [27]: #Reviewing the first 10 rows of the data set
        df.head(10)
```

Out[27]:

	RequestId	Problem Code	Date Initiated	Description	Priority	Category	Submit To	Dispatch To	Address	Zone	Council District
0	218033	Tree Removal	2025-01-01 03:08 PM	Tree Removal Inspection	0.0	TREEMAINTENANCE	UF - CVM, VOICEMAIL	UF - ERICKSON, IAN	5151 Hull St, Houston, Texas, 77021	6, ZONE	D
1	218035	Tree Removal	2025-01-02 08:47 AM	Tree Removal Inspection	3.0	TREEMAINTENANCE	UF - CVM, VOICEMAIL	UF - ERICKSON, IAN	3011 N Macgregor Way, Houston, Texas, 77004	6, ZONE	D
2	218037	Tree Trim	2025-01-02 09:38 AM	Tree Trim Inspection	0.0	TREEMAINTENANCE	UF - CDY, D365	UF - ERICKSON, IAN	5524 Gessner Dr, Houston, Texas, 77041	3, ZONE	A
3	218038	Tree Trim	2025-01-02 09:50 AM	Tree Trim Inspection	0.0	TREEMAINTENANCE	UF - NAS, NO ANSWER	UF - CASTILLO, ALEJANDRO	5131 Idaho St, Houston, Texas, 77021	6, ZONE	D
4	218040	Tree Removal	2025-01-02 10:26 AM	Tree Removal Inspection	0.0	TREEMAINTENANCE	UF - CPH, PHONE	UF - LEIJA, DANIEL	3603 Freshmeadows Dr, Houston, Texas, 77063	2, ZONE	F
5	218048	Tree Trim	2025-01-02 11:32 AM	Tree Trim Inspection	0.0	TREEMAINTENANCE	UF - CVM, VOICEMAIL	UF - CASTILLO, ALEJANDRO	10057 Bob White Dr, Houston, Texas, 77096	7, ZONE	C
6	218052	Tree Removal	2025-01-02 04:15 PM	Tree Removal Inspection	0.0	TREEMAINTENANCE	UF - CDY, D365	UF - CASTILLO, ALEJANDRO	6014 Willowbend Blvd, Houston, Texas, 77096	7, ZONE	C
7	218053	Tree Removal	2025-01-02 09:28 PM	Tree Removal Inspection	0.0	TREEMAINTENANCE	UF - NCI, NO CONTACT INFO	UF - LEIJA, DANIEL	1504 Upland Glen Way, Houston, Texas, 77043	3, ZONE	A
8	218056	Tree Stump Grinding	2025-01-03 08:21 AM	Tree Stump Inspection	3.0	TREEMAINTENANCE	UF - CIP, IN-PERSON	UF - ERICKSON, IAN	7014 Vandeman St, Houston, Texas, 77087	5, ZONE	I

	RequestId	Problem Code	Date Initiated	Description	Priority	Category	Submit To	Dispatch To	Address	Zone	Council District
9	218061	Tree Trim	2025-01-03 09:54 AM	Tree Trim Inspection	0.0	TREEMAINTENANCE	UF - CPH, PHONE	UF - LEIJA, DANIEL	6001 Fannin St, Houston, Texas, 77030 (Herm...	7, ZONE	D

```
In [28]: #Standardizing column headings to remove spaces and make all lower case
df.columns = (df.columns.str.replace(' ', '_').str.lower().str.strip())
df
```

Out[28]:

	requestid	problem_code	date_initiated	description	priority	category	submit_to	dispatch_to	address	z
0	218033	Tree Removal	2025-01-01 03:08 PM	Tree Removal Inspection	0.0	TREEMAINTENANCE	UF - CVM, VOICEMAIL	UF - ERICKSON, IAN	5151 Hull St, Houston, Texas, 77021	ZC
1	218035	Tree Removal	2025-01-02 08:47 AM	Tree Removal Inspection	3.0	TREEMAINTENANCE	UF - CVM, VOICEMAIL	UF - ERICKSON, IAN	3011 N Macgregor Way, Houston, Texas, 77004	ZC
2	218037	Tree Trim	2025-01-02 09:38 AM	Tree Trim Inspection	0.0	TREEMAINTENANCE	UF - CDY, D365	UF - ERICKSON, IAN	5524 Gessner Dr, Houston, Texas, 77041	ZC
3	218038	Tree Trim	2025-01-02 09:50 AM	Tree Trim Inspection	0.0	TREEMAINTENANCE	UF - NAS, NO ANSWER	UF - CASTILLO, ALEJANDRO	5131 Idaho St, Houston, Texas, 77021	ZC
4	218040	Tree Removal	2025-01-02 10:26 AM	Tree Removal Inspection	0.0	TREEMAINTENANCE	UF - CPH, PHONE	UF - LEIJA, DANIEL	3603 Freshmeadows Dr, Houston, Texas, 77063	ZC
...	...	...	...	...	...	...	...	...	...	...
5888	230088	Tree Trim	2025-12-31 10:32 AM	Tree Trim Inspection	0.0	TREEMAINTENANCE	UF - CIP, IN- PERSON	UF - CASTILLO, ALEJANDRO	5930 Densmore Dr, Houston, Texas, 77035	ZC
5889	230091	Tree Removal	2025-12-31 11:38 AM	Tree Removal Inspection	1.0	TREEMAINTENANCE	UF - CIP, IN- PERSON	UF - LEIJA, DANIEL	6330 Jim St, Houston, Texas, 77092	ZC
5890	230092	Tree Trim	2025-12-31 11:43 AM	Tree Trim Inspection	0.0	TREEMAINTENANCE	UF - NCI, NO CONTACT INFO	UF - LEIJA, DANIEL	8923 Winding River Dr, Houston, Texas, 77088	ZC
5891	230094	Tree Removal	2025-12-31 01:49 PM	Tree Removal Inspection	0.0	TREEMAINTENANCE	UF - CIP, IN- PERSON	UF - LEIJA, DANIEL	7622 Richmond Ave, Houston, Texas, 77063	ZC
5892	230096	Tree Removal	2025-12-31 02:44 PM	Tree Removal Inspection	0.0	TREEMAINTENANCE	UF - NAS, NO ANSWER	UF - JONES, CODY	7001 Corporate Dr, Houston, Texas, 77036	ZC

5893 rows × 14 columns

```
In [29]: #Dropping unnecessary columns - priority, submit_to, council_district  
df = df.drop(columns = ['priority', 'submit_to', 'council_district'])
```

```
In [30]: #Filtering out all null values from zone  
df['zone'].isnull()
```

```
Out[30]: 0      False  
1      False  
2      False  
3      False  
4      False  
      ...  
5888   False  
5889   False  
5890   False  
5891   False  
5892   False  
Name: zone, Length: 5893, dtype: bool
```

```
In [31]: #Displaying the filter as a dataframe to determine the number of rows with null values in zone  
df[df['zone'].isnull()]
```

Out[31]:

	requestid	problem_code	date_initiated	description	category	dispatch_to	address	zone	keymap	zip_code	
331	219041	Tree Code Violation	2025-02-10 08:15 AM	Tree Code Violation Inspection	TREEPROTECTION	UF - CASTILLO, ALEJANDRO	Cullen Blvd & Kilnar, Houston, Texas, 77047...	NaN	613D	77047	(C)
827	220292	Tree Removal	2025-03-24 12:08 AM	Tree Removal Inspection	TREEMAINTENANCE	UF - ERICKSON, IAN	12300 Aldine Westfield Rd, Houston, Texas, ...	NaN	413M	77093	(C)
828	220296	Tree Removal	2025-03-24 07:51 AM	Tree Removal Inspection	TREEMAINTENANCE	UF - CASTILLO, ALEJANDRO	Scarsdale Blvd & Beamer Rd, Houston, Texas, ...	NaN	616G	77089	(C)
949	220623	Tree Removal	2025-03-31 01:18 PM	Tree Removal Inspection	TREEMAINTENANCE	UF - ERICKSON, IAN	5210 Aldine Bender Rd, Houston, Texas, 77032	NaN	374Z	77032	(C)
1047	220836	Tree Trim	2025-04-04 02:20 PM	Tree Trim Inspection	TREEMAINTENANCE	UF - ERICKSON, IAN	1324 Pine Chase Dr, Houston, Texas, 77055	NaN	451W	77055	(C)
1372	221530	Tree Removal	2025-04-23 07:16 PM	Tree Removal Inspection	TREEMAINTENANCE	UF - ERICKSON, IAN	Deussen Pkwy & Deussen Dr, Houston, Texas, ...	NaN	417H	77044	(C)
1556	221874	Tree Trim	2025-05-02 04:21 PM	Tree Trim Inspection	TREEMAINTENANCE	UF - ERICKSON, IAN	Waycreek Rd & Kuykendahl Rd, Houston, Texas,...	NaN	331Q	77090	(C)

	requestid	problem_code	date_initiated	description	category	dispatch_to	address	zone	keymap	zip_code
1746	222227	Tree Removal	2025-05-13 01:12 PM	Tree Removal Inspection		NaN	NaN	NaN	NaN	NaN
2051	222804	Tree Removal	2025-05-28 06:57 PM	Tree Removal Inspection		NaN	NaN	NaN	NaN	NaN
3955	226066	Tree Removal	2025-08-22 01:16 AM	Tree Removal Inspection	TREEMAINTENANCE	UF - LEIJA, DANIEL	FM-1960 & Terrace Oaks Dr, Houston, Texas, ...	NaN	331T	77068 C
4110	226376	Tree Trim	2025-08-29 06:53 PM	Tree Trim Inspection		NaN	NaN	NaN	NaN	NaN
4504	227101	Tree Trim	2025-09-17 02:38 PM	Tree Trim Inspection	TREEMAINTENANCE	UF - CASTILLO, ALEJANDRO	Beechnut St & N Braeswood Blvd, Houston, Tex...	NaN	531R	77096 C

In [32]: *#Replacing null values in zone with known values*

```
df.iloc[331, 7] = '6, ZONE'
df.iloc[827, 7] = '4, ZONE'
df.iloc[828, 7] = '6, ZONE'
df.iloc[949, 7] = '4, ZONE'
df.iloc[1047, 7] = '3, ZONE'
df.iloc[1372, 7] = '4, ZONE'
df.iloc[1556, 7] = '4, ZONE'
df.iloc[3955, 7] = '4, ZONE'
df.iloc[4504, 7] = '7, ZONE'
```

In [33]: *#Confirming null values in zone have been replaced with the known values*

```
df[df['zone'].isnull()]
```

Out[33]:

	requestid	problem_code	date_initiated	description	category	dispatch_to	address	zone	keymap	zip_code	status
1746	222227	Tree Removal	2025-05-13 01:12 PM	Tree Removal Inspection	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2051	222804	Tree Removal	2025-05-28 06:57 PM	Tree Removal Inspection	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4110	226376	Tree Trim	2025-08-29 06:53 PM	Tree Trim Inspection	NaN	NaN	NaN	NaN	NaN	NaN	NaN

In [34]: *#Filtering out all null values from category*
`df['category'].isnull()`

Out[34]:

0	False
1	False
2	False
3	False
4	False
...	
5888	False
5889	False
5890	False
5891	False
5892	False

Name: category, Length: 5893, dtype: bool

In [35]: *#Displaying the filter as a dataframe to determine the number of rows with null values in category*
`df[df['category'].isnull()]`

Out[35]:

	requestid	problem_code	date_initiated	description	category	dispatch_to	address	zone	keymap	zip_code	status
1746	222227	Tree Removal	2025-05-13 01:12 PM	Tree Removal Inspection	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2051	222804	Tree Removal	2025-05-28 06:57 PM	Tree Removal Inspection	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4110	226376	Tree Trim	2025-08-29 06:53 PM	Tree Trim Inspection	NaN	NaN	NaN	NaN	NaN	NaN	NaN

```
In [36]: #Replacing null values in category with known values
df.iloc[1746, 4] = 'TREEMAINTENANCE'
df.iloc[2051, 4] = 'TREEMAINTENANCE'
df.iloc[4110, 4] = 'TREEMAINTENANCE'
```

```
In [38]: #Confirming null value in category has been replaced with the known value
df[df['category'].isnull()]
```

```
Out[38]:  requestid  problem_code  date_initiated  description  category  dispatch_to  address  zone  keymap  zip_code  status
```

```
In [39]: #Calculating the sum of all missing values for each column to confirm again that all missing values have
df.isna().sum()
```

```
Out[39]: requestid      0
problem_code    0
date_initiated  0
description     0
category        0
dispatch_to     4
address         3
zone           3
keymap          3
zip_code        4
status          3
dtype: int64
```

```
In [40]: #Replacing null values in status with known values
df.iloc[1746, 10] = 'CLOSED'
df.iloc[2051, 10] = 'CLOSED'
df.iloc[4110, 10] = 'CLOSED'
```

```
In [41]: #Filtering out all null values from dispatch_to
df['dispatch_to'].isnull()
```

```
Out[41]: 0      False
         1      False
         2      False
         3      False
         4      False
         ...
        5888    False
        5889    False
        5890    False
        5891    False
        5892    False
        Name: dispatch_to, Length: 5893, dtype: bool
```

```
In [42]: #Displaying the filter as a dataframe to determine the number of rows with null values in dispatch_to
df[df['dispatch_to'].isnull()]
```

```
Out[42]:
```

	requestid	problem_code	date_initiated	description	category	dispatch_to	address	zone	keymap	zip_code	
1746	222227	Tree Removal	2025-05-13 01:12 PM	Tree Removal Inspection	TREEMAINTENANCE	NaN	NaN	NaN	NaN	NaN	CL
2051	222804	Tree Removal	2025-05-28 06:57 PM	Tree Removal Inspection	TREEMAINTENANCE	NaN	NaN	NaN	NaN	NaN	CL
2848	224222	Tree Trim	2025-07-03 09:44 AM	Tree Trim Inspection	TREEMAINTENANCE	NaN	2304 Woodland Park Dr, Houston, Texas, 77077	2, ZONE	489T	77077	
4110	226376	Tree Trim	2025-08-29 06:53 PM	Tree Trim Inspection	TREEMAINTENANCE	NaN	NaN	NaN	NaN	NaN	CL

```
In [43]: #Replacing null value in dispatch_to with known value
df.iloc[2848, 5] = 'UF - JONES, CODY'
```

```
In [44]: #Splitting 'date_initiated' field into two fields to separate date and time
df[['initiated_date', 'initiated_time']] = df['date_initiated'].str.split(' ', expand=True, n=1)
df
```

Out[44]:

	requestid	problem_code	date_initiated	description	category	dispatch_to	address	zone	keymap	zip_coc
0	218033	Tree Removal	2025-01-01 03:08 PM	Tree Removal Inspection	TREEMAINTENANCE	UF - ERICKSON, IAN	5151 Hull St, Houston, Texas, 77021	6, ZONE	534N	7702
1	218035	Tree Removal	2025-01-02 08:47 AM	Tree Removal Inspection	TREEMAINTENANCE	UF - ERICKSON, IAN	3011 N Macgregor Way, Houston, Texas, 77004	6, ZONE	533G	7700
2	218037	Tree Trim	2025-01-02 09:38 AM	Tree Trim Inspection	TREEMAINTENANCE	UF - ERICKSON, IAN	5524 Gessner Dr, Houston, Texas, 77041	3, ZONE	450A	7704
3	218038	Tree Trim	2025-01-02 09:50 AM	Tree Trim Inspection	TREEMAINTENANCE	UF - CASTILLO, ALEJANDRO	5131 Idaho St, Houston, Texas, 77021	6, ZONE	534N	7702
4	218040	Tree Removal	2025-01-02 10:26 AM	Tree Removal Inspection	TREEMAINTENANCE	UF - LEIJA, DANIEL	3603 Freshmeadows Dr, Houston, Texas, 77063	2, ZONE	490Z	7706
...	...	...	...	...	...	...	...	...	...	...
5888	230088	Tree Trim	2025-12-31 10:32 AM	Tree Trim Inspection	TREEMAINTENANCE	UF - CASTILLO, ALEJANDRO	5930 Densmore Dr, Houston, Texas, 77035	7, ZONE	571E	7703
5889	230091	Tree Removal	2025-12-31 11:38 AM	Tree Removal Inspection	TREEMAINTENANCE	UF - LEIJA, DANIEL	6330 Jim St, Houston, Texas, 77092	3, ZONE	451K	7709
5890	230092	Tree Trim	2025-12-31 11:43 AM	Tree Trim Inspection	TREEMAINTENANCE	UF - LEIJA, DANIEL	8923 Winding River Dr, Houston, Texas, 77088	3, ZONE	412L	7708
5891	230094	Tree Removal	2025-12-31 01:49 PM	Tree Removal Inspection	TREEMAINTENANCE	UF - LEIJA, DANIEL	7622 Richmond Ave, Houston, Texas, 77063	2, ZONE	490Z	7706
5892	230096	Tree Removal	2025-12-31 02:44 PM	Tree Removal Inspection	TREEMAINTENANCE	UF - JONES, CODY	7001 Corporate Dr, Houston, Texas, 77036	2, ZONE	529H	7703

5893 rows × 13 columns

```
In [45]: #Converting 'date_initiated' field to datetime data type  
df['initiated_date'] = pd.to_datetime(df['initiated_date'], dayfirst=False)  
df
```

Out[45]:

	requestid	problem_code	date_initiated	description	category	dispatch_to	address	zone	keymap	zip_coc
0	218033	Tree Removal	2025-01-01 03:08 PM	Tree Removal Inspection	TREEMAINTENANCE	UF - ERICKSON, IAN	5151 Hull St, Houston, Texas, 77021	6, ZONE	534N	7702
1	218035	Tree Removal	2025-01-02 08:47 AM	Tree Removal Inspection	TREEMAINTENANCE	UF - ERICKSON, IAN	3011 N Macgregor Way, Houston, Texas, 77004	6, ZONE	533G	7700
2	218037	Tree Trim	2025-01-02 09:38 AM	Tree Trim Inspection	TREEMAINTENANCE	UF - ERICKSON, IAN	5524 Gessner Dr, Houston, Texas, 77041	3, ZONE	450A	7704
3	218038	Tree Trim	2025-01-02 09:50 AM	Tree Trim Inspection	TREEMAINTENANCE	UF - CASTILLO, ALEJANDRO	5131 Idaho St, Houston, Texas, 77021	6, ZONE	534N	7702
4	218040	Tree Removal	2025-01-02 10:26 AM	Tree Removal Inspection	TREEMAINTENANCE	UF - LEIJA, DANIEL	3603 Freshmeadows Dr, Houston, Texas, 77063	2, ZONE	490Z	7706
...	...	...	...	...	...	...	...	...	...	...
5888	230088	Tree Trim	2025-12-31 10:32 AM	Tree Trim Inspection	TREEMAINTENANCE	UF - CASTILLO, ALEJANDRO	5930 Densmore Dr, Houston, Texas, 77035	7, ZONE	571E	7703
5889	230091	Tree Removal	2025-12-31 11:38 AM	Tree Removal Inspection	TREEMAINTENANCE	UF - LEIJA, DANIEL	6330 Jim St, Houston, Texas, 77092	3, ZONE	451K	7709
5890	230092	Tree Trim	2025-12-31 11:43 AM	Tree Trim Inspection	TREEMAINTENANCE	UF - LEIJA, DANIEL	8923 Winding River Dr, Houston, Texas, 77088	3, ZONE	412L	7708
5891	230094	Tree Removal	2025-12-31 01:49 PM	Tree Removal Inspection	TREEMAINTENANCE	UF - LEIJA, DANIEL	7622 Richmond Ave, Houston, Texas, 77063	2, ZONE	490Z	7706
5892	230096	Tree Removal	2025-12-31 02:44 PM	Tree Removal Inspection	TREEMAINTENANCE	UF - JONES, CODY	7001 Corporate Dr, Houston, Texas, 77036	2, ZONE	529H	7703

5893 rows × 13 columns

```
In [46]: #Verifying data types for all fields
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5893 entries, 0 to 5892
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   requestid             5893 non-null   int64
1   problem_code          5893 non-null   object
2   date_initiated        5893 non-null   object
3   description           5893 non-null   object
4   category              5893 non-null   object
5   dispatch_to          5890 non-null   object
6   address               5890 non-null   object
7   zone                  5890 non-null   object
8   keymap                5890 non-null   object
9   zip_code              5889 non-null   object
10  status                5893 non-null   object
11  initiated_date         5893 non-null   datetime64[ns]
12  initiated_time         5893 non-null   object
dtypes: datetime64[ns](1), int64(1), object(11)
memory usage: 598.6+ KB
```

```
In [47]: #Saving cleaned data set as new csv file
df.to_csv('My_Projects/Urban_Forestry/Cityworks_SR/Cityworks_SRs_2025_cleaned.csv', index=False)
```

```
In [ ]:
```